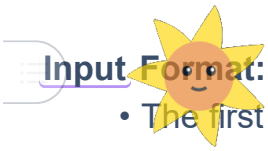
1.1.12. Longest C... 06:56

Write a C program that takes three strings X, Y, and Z as input and finds the length of the Longest Common Subsequence (LCS) that is present in all three strings.



Input Format:

- The first line contains the string X.
- The second line contains the string Y.
- The third line contains the string Z.

Output Format:

- A single integer representing the length of the Longest Common Subsequence among the three given strings.

Constraints:

- $1 \leq length\ of\ X \leq 10^2$
- $1 \leq length\ of\ Y \leq 10^2$
- $1 \leq length\ of\ Z \leq 10^2$

Sample Test Case:

Input:

```
Twenty
Thirty
Forty
```

Output:

```
2
```

Explanation:

- The Longest common sub-sequence in the given three strings is "**ty**" so the answer is 2 because it is having length of 2.

Sample Test Cases +

Explorer

C lcss.c

Submit

Debugger

```c
#include<stdio.h>
#include<string.h>
#include<stdbool.h>
#define MAX_LENGTH 100

int max(int a, int b) {if(a>b){return a;}else{return b;}}

int lcsOf3(char X[], char Y[], char Z[], int m, int n, int o){

    //write your code here...
    if (m > 0 && X[m - 1] == '\n') m--;
    if (n > 0 && Y[n - 1] == '\n') n--;
    if (o > 0 && Z[o - 1] == '\n') o--;

    int dp[m + 1][n + 1][o + 1];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            for (int k = 0; k <= o; k++) {
                if (i == 0 || j == 0 || k == 0) {

dp[i][j][k] = 0;
                } else if (X[i - 1] == Y[j - 1] && X[i - 1] == Z[k - 1]) {

```

‹ Prev Reset Submit Next ›

```
[j - 1][k - 1] + 1;
                } else {

dp[i][j][k] =
max(max(dp[i - 1][j]
[k], dp[i][j - 1][k]),
dp[i][j][k - 1]);
                }
            }
        }
    }
    return dp[m][n][o];


}

int main()
{   char x[MAX_LENGTH],
y[MAX_LENGTH],z[MAX_LENG
TH];
    fgets(x,
MAX_LENGTH, stdin);
    fgets(y,
MAX_LENGTH, stdin);
    fgets(z,
MAX_LENGTH, stdin);
    printf("%d",
lcsOf3(x, y, z,
strlen(x), strlen(y),
strlen(z)));

    return 0;
}
```